Apr. 28

- Spent ~45 mins. on Primary source project
  - Will spend ~30 mins. each class on it

- Ideals are subrings + mult. absorption property

- Def. $I \subseteq R$ is an ideal if I is a subring such that ar and ra are in I whenever $a \in I$ and $r \in R$

Quotient Rings

- Let R be a ring and I an ideal of R. Write $R/I = \{r+I : r \in R\}$ for the set of cosets of I in R. ($r+I = \{r+a : a \in I\}$)
  - Define addition $(r+I)+(s+I) = (r+s)+I \ \forall r,s \in R$. We already know that $(R/I, +)$ is an Abelian group
  - Define multiplication $(r+I)(s+I) = rs+I \ \forall r,s \in R$

- Thm. 14.2 R/I is a ring. In fact, if A is a subring of R, R/A is a ring if and only if A is an ideal. If A is not an ideal, R/A is only a group, not ring
  - Proof

    from our work with groups, we know $(R/I, +)$ is an Abelian group. ★

    If $r+I, s+I, t+I$ are in R/I, we check $((r+I)(s+I))(t+I) = (rs+I)(t+I)$

    $= (rs)t+I = r(st)+I = (r+I)(st+I) = (r+I)((s+I)(t+I))$. ★

    ($\cdot$ is associative in R)

    Left and Right distributivity are also inherited from R. ★

    To show: mult. is well-defined, suppose $r+I = r'+I$ and $s+I = s'+I$. We want $(r+I)(s+I) = rs+I$ is the same set as $(r'+I)(s'+I) = r's'+I$.

    Since $r+I = r'+I$ and $s+I = s'+I$, we have $r' = r+a$ and $s' = s+b$ for some $a,b \in I$. Then $r's' = (r+a)(s+b) = rs+as+rb+ab$. Since I is an ideal and $a,b \in I$, $as \in I$ and $rb \in I$ (mult. absorption) thus $as+rb+ab \in I$. Hence $r's'+I = rs+I$.

Ring conditions:
- $+$ ✓, $\cdot$ ✓ well defined
- Closure under $+$ ✓, $\cdot$ ✓
- $(R/I, +)$ ✓ Abelian group
- Associative $\cdot$ ✓
- L & R Distributivity ✓

★ Cosets are equal if they differ by an element of the group/ring being quotiented by

Ex. $1+4\mathbb{Z} = 81+4\mathbb{Z}$ $\quad 1 = 81-80$ and $80 \in 4\mathbb{Z}$, representatives' difference is an element of the group/ring that is quotiented by, here $4\mathbb{Z}$